Number 6 — Direct and inverse proportion · Fractional and negative indices

No calculator for questions 4 and 5. Marks in brackets. ★ = harder question.

1 The cost of copper wire is directly proportional to its length. A 2.5 m length costs £4.50. [4]

a Find the cost of 7 m of wire. **b** What length can be bought for £27?

c Write a formula connecting the cost C (in £) and the length L (in m).

2 The time taken for a journey is inversely proportional to the average speed. At an average speed of 60 km/h the journey takes 40 minutes. [4]

a How long does it take at 80 km/h?

b What average speed is needed to complete it in 32 minutes?

3 A tank can be emptied by 4 identical pumps in 9 hours. The time taken is inversely proportional to the number of pumps working. [4]

a How long would 6 pumps take?

b How many pumps are needed to empty the tank in 4.5 hours?

c Sketch the graph of time against number of pumps.

4 Work out, without a calculator. Give exact answers. [9]

a $49^{\frac{1}{2}}$ **b** $27^{\frac{2}{3}}$ **c** $16^{-\frac{1}{2}}$ **d** $\left(\frac{25}{9}\right)^{\frac{3}{2}}$ **e** 5^{-2}

f $\left(\frac{2}{3}\right)^{-3}$ **g** $8^{-\frac{2}{3}}$ **h** $\left(1\frac{7}{9}\right)^{\frac{1}{2}}$ **i** $1 \div 81^{-\frac{1}{4}}$

5 Solve for x, without a calculator. [4]

a $\left(\frac{4}{9}\right)^{x} = \frac{27}{8}$ **b** $125^{x} = \frac{1}{25}$

Total: 25 marks

Check: direct proportion — find the value for 1 unit (or keep the ratio constant) · inverse proportion — the product is constant · $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$: root first · $a^{-n} = \frac{1}{a^n}$ and $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ · to solve $a^{f(x)} = a^{g(x)}$, equate the indices.